

Computer Networks

Address Resolution Protocol (ARP)

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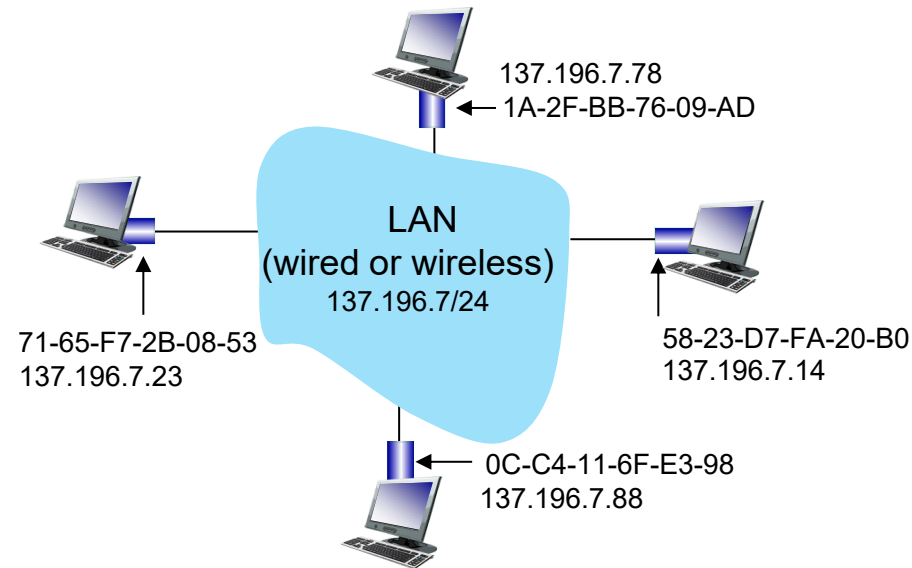
IP address vs MAC address

- 32-bit IP address:
 - Network-layer address for interface
 - Used for layer 3 (network layer) forwarding
 - e.g.: 128.119.40.136
 - MAC (or LAN or physical or Ethernet) address:
 - Used “locally” to get frame from one interface to another physically-connected interface (same subnet, in IP-addressing sense)
 - 48-bit MAC address (for most LANs) burned in NIC ROM, also sometimes software settable
 - e.g.: 1A-2F-BB-76-09-AD
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IP address vs MAC addresses

Each interface on LAN

- has unique 48-bit **MAC** address
- has a locally unique 32-bit IP address



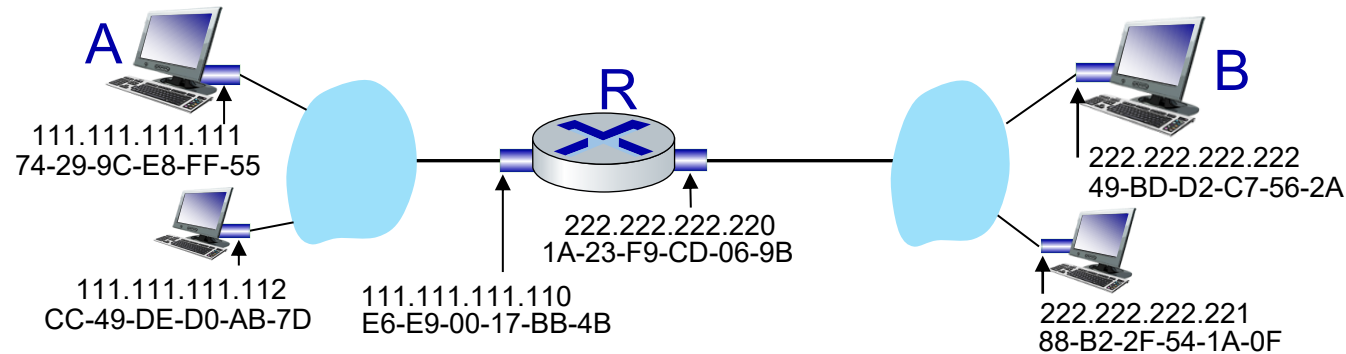
MAC addresses

- MAC address allocation administered by IEEE
 - Manufacturer buys portion of MAC address space (to assure uniqueness)
 - MAC address is flat and portable
 - can move interface from one LAN to another
 - IP address is not portable: depends on IP subnet to which node is attached
 - Analogy:
 - MAC address: like Aadhar Card Number
 - IP address: like postal address
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ARP: Address Resolution Protocol

Suppose we want to send a datagram from A to B via R

- The packet will go through multiple L2 switches
- What does A need to know?
 - A need to know B's IP address
 - A need to know IP address of first hop router, R
 - A need to know R's MAC address



ARP: Address Resolution Protocol

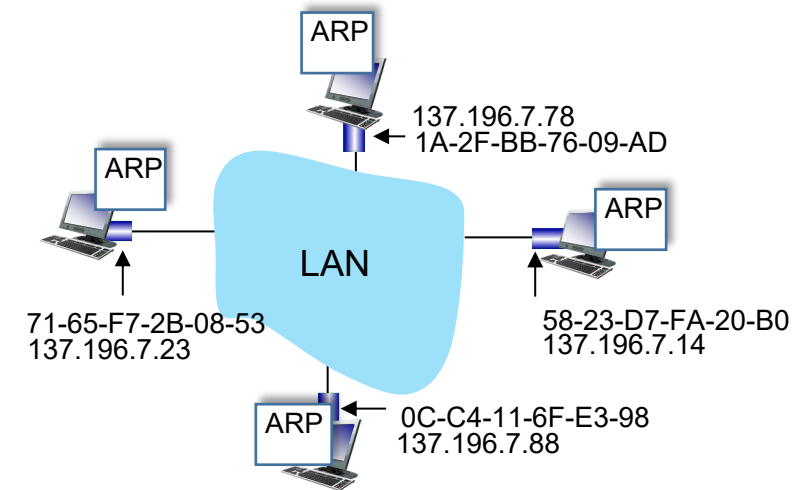
How to determine interface's MAC address, knowing its IP address?

ARP table: each IP node (host, router) on LAN has table

- IP/MAC address mappings for some LAN nodes:

< IP address; MAC address; TTL >

- TTL (Time To Live): time after which address mapping will be forgotten (typically 20 min)



ARP: Address Resolution Protocol

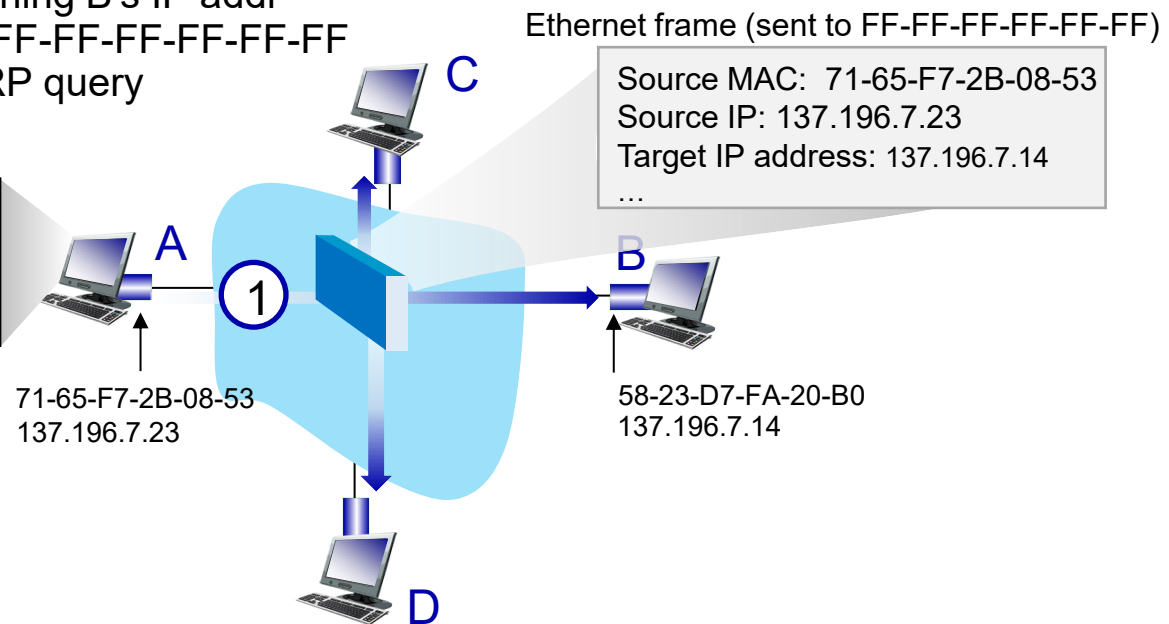
Example: A wants to send datagram to B

- B's MAC address not in A's ARP table, so A uses ARP to find B's MAC address

- A broadcasts ARP query, containing B's IP addr
- 1 destination MAC address = FF-FF-FF-FF-FF-FF
 - all nodes on LAN receive ARP query

ARP table in A

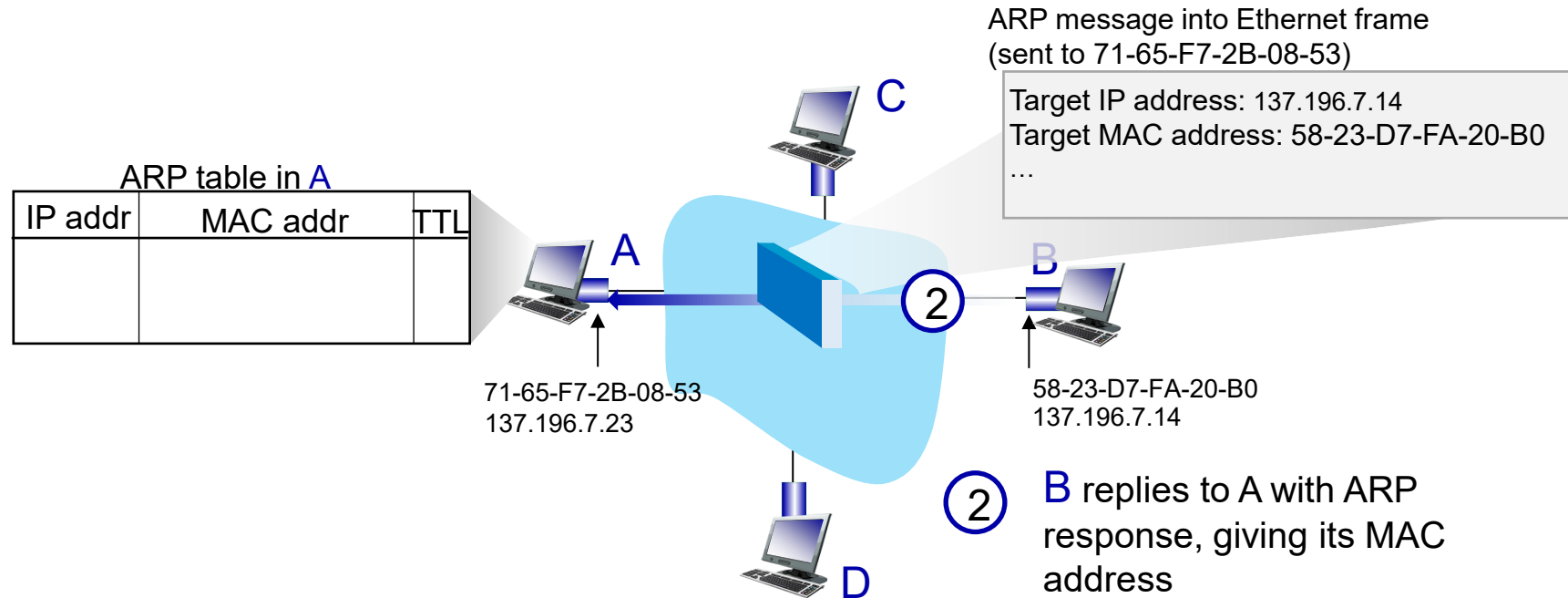
IP addr	MAC addr	TTL



ARP: Address Resolution Protocol

Example: A wants to send datagram to B

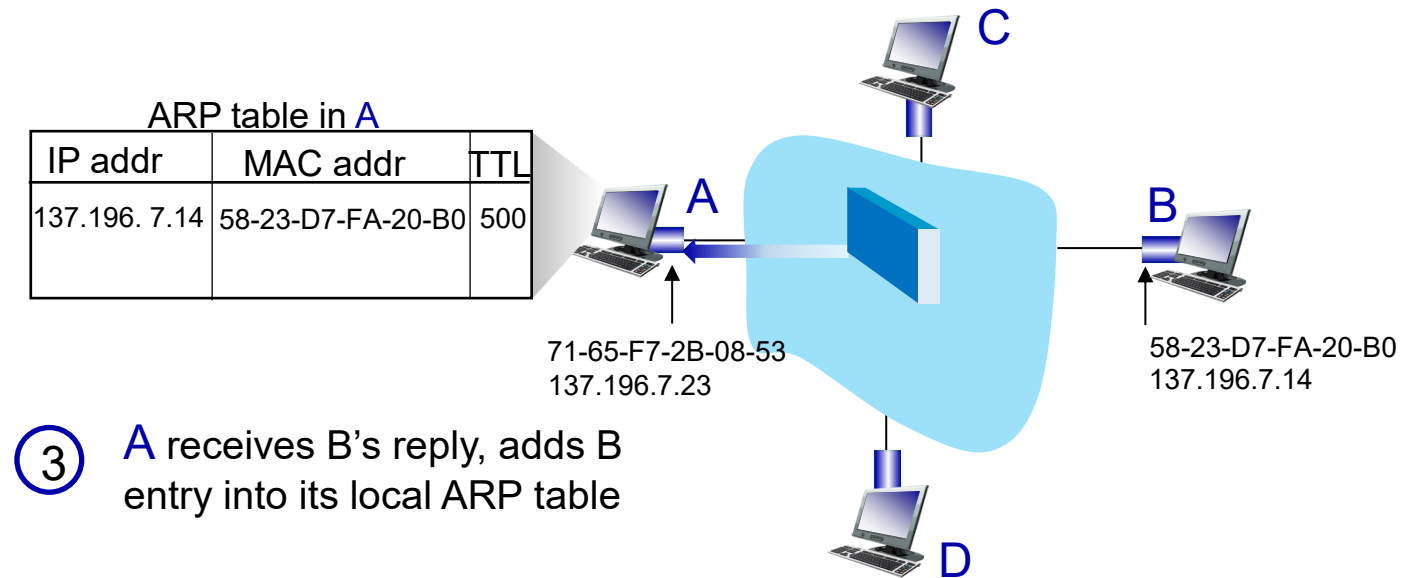
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ARP: Address Resolution Protocol

Example: A wants to send datagram to B

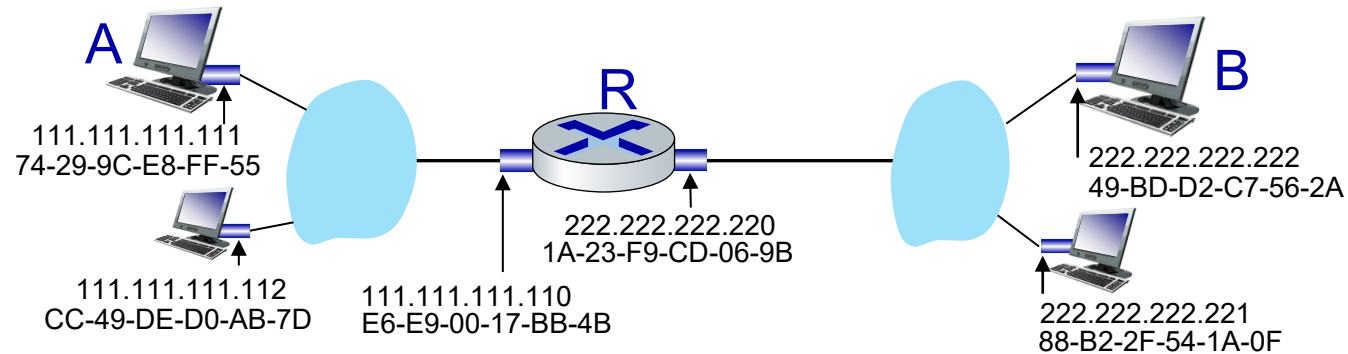
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Routing to Another Subnet

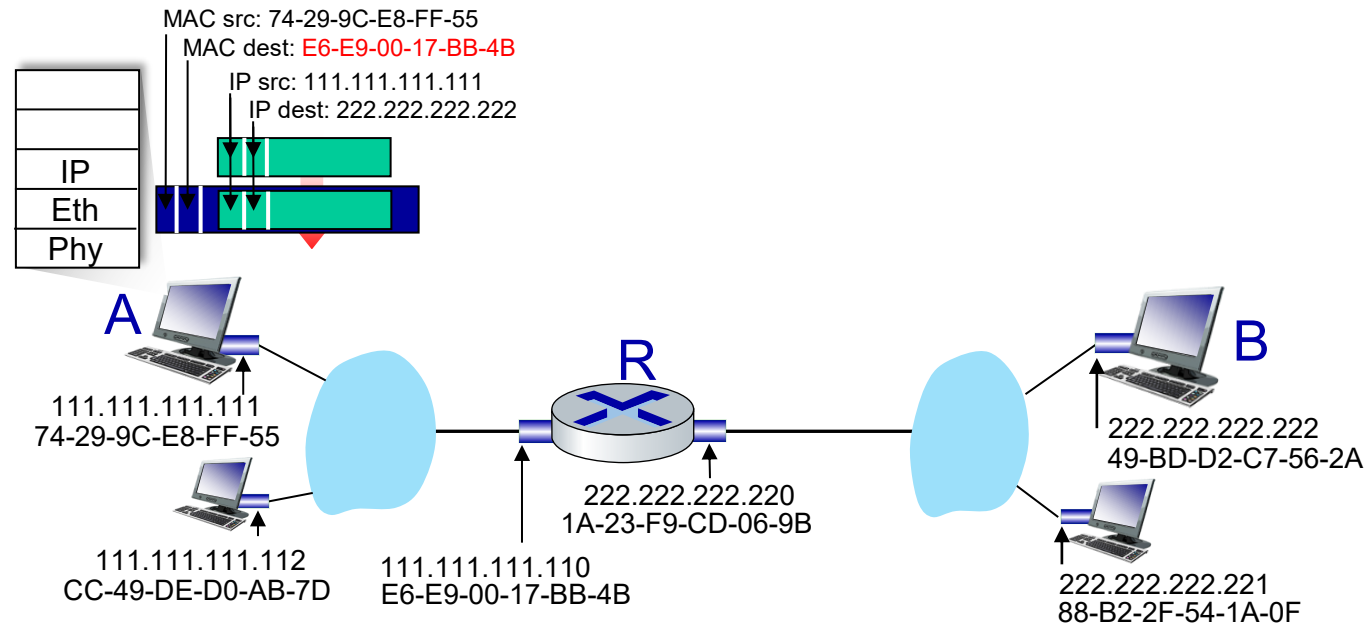
Walkthrough: sending a datagram from A to B via R

- Focus on addressing – at IP (datagram) and MAC layer (frame) levels
- Assume that:
 - A knows B's IP address
 - A knows IP address of first hop router, R
 - A knows R's MAC address



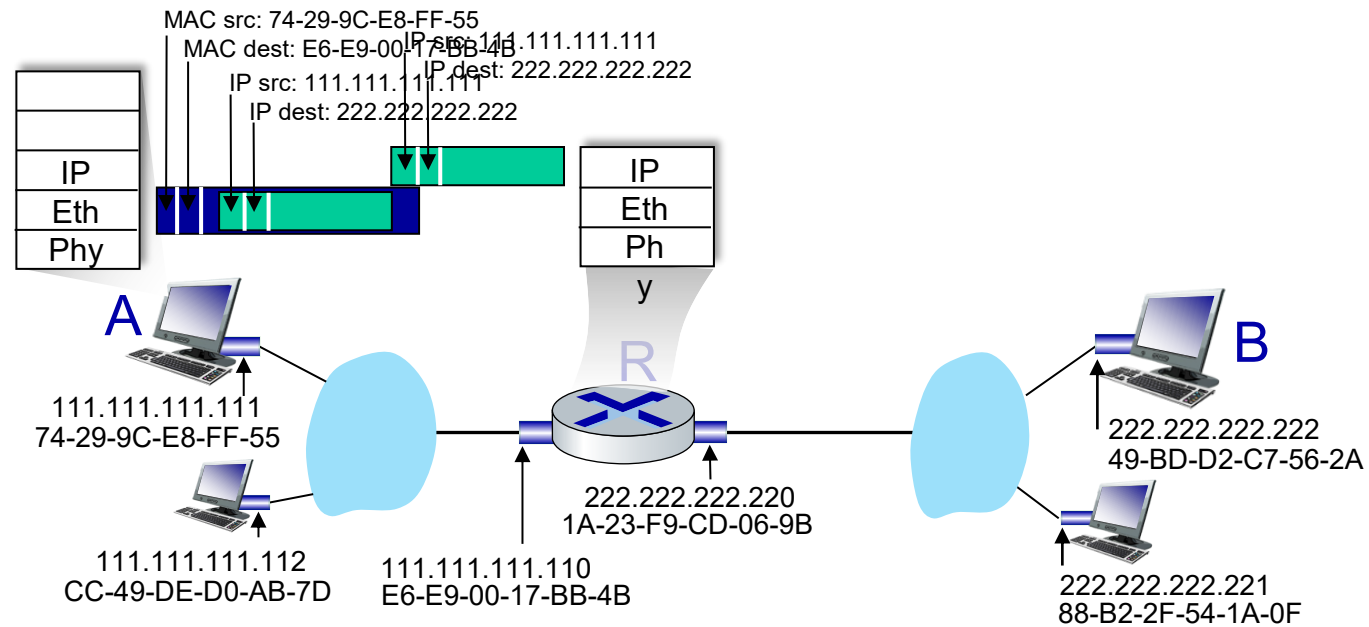
Routing to Another Subnet

- A creates IP datagram with IP source A, destination B
- A creates link-layer frame containing A-to-B IP datagram
 - **R's** MAC address is frame's destination



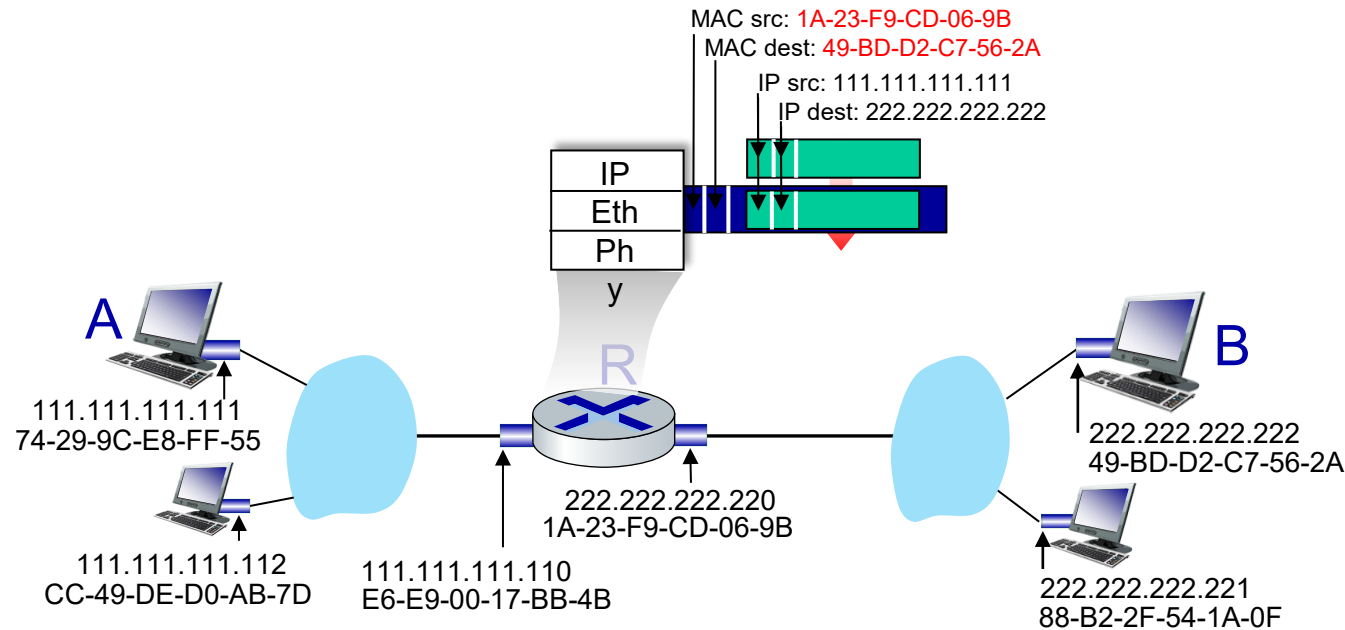
Routing to Another Subnet

- Frame sent from A to R
- Frame received at R, datagram removed, passed up to IP



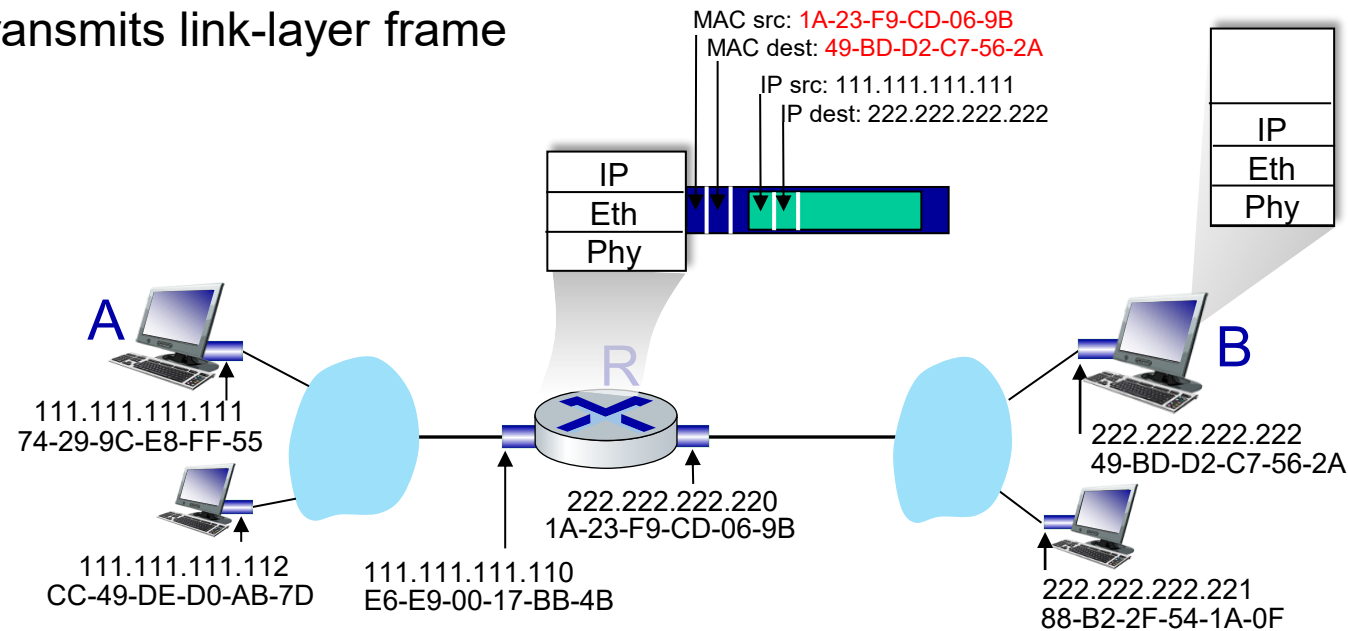
Routing to Another Subnet

- R determines outgoing interface, passes datagram with IP source A, destination B to link layer
- R creates link-layer frame containing A-to-B IP datagram. Frame destination address: B's MAC address



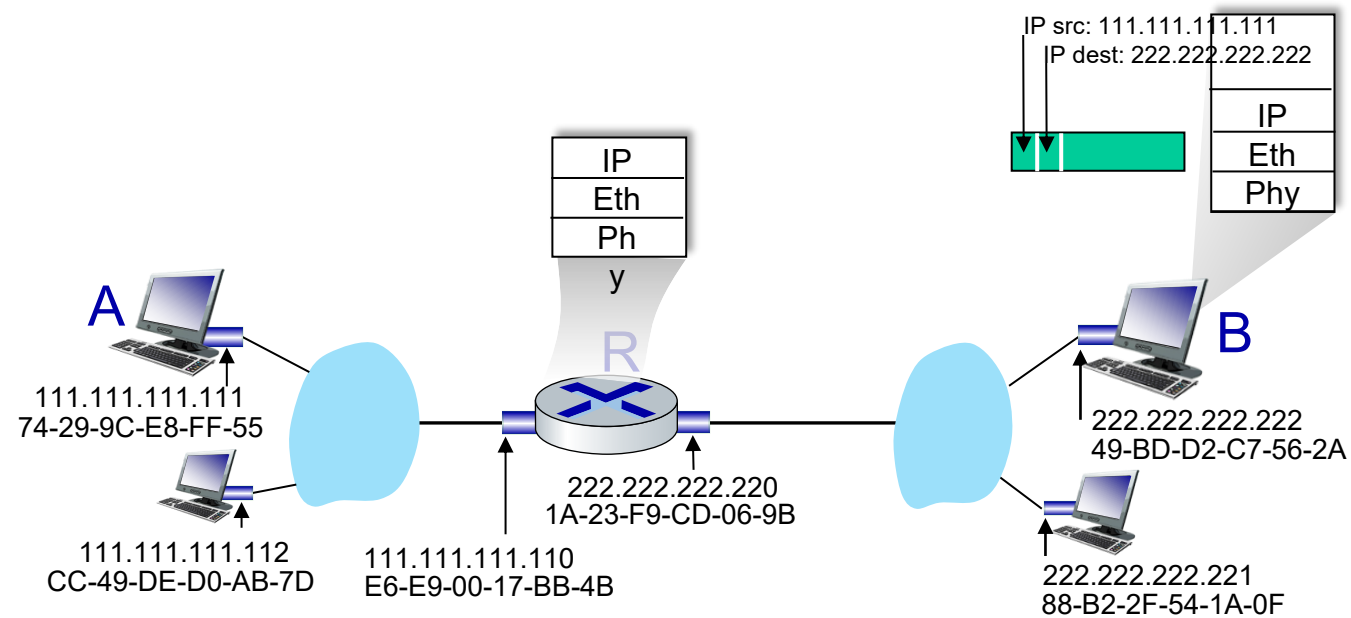
Routing to Another Subnet

- R determines outgoing interface, passes datagram with IP source A, destination B to link layer
- R creates link-layer frame containing A-to-B IP datagram. Frame destination address: B's MAC address
- Transmits link-layer frame



Routing to Another Subnet

- B receives frame, extracts IP datagram destination B
- B passes datagram up protocol stack to IP



Summary

- ❑ MAC address vs IP address:
 - ❑ Address resolution protocol (ARP)
 - MAC address from IP address
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